

Problem

Two loads connected in parallel draw a total of 2.4 kW at 0.8 pf lagging from a 120-V rms, 60-Hz line. One load absorbs 1.5 kW at a 0.707 pf lagging. Determine: (a) the pf of the second load, (b) the parallel element required to correct the pf to 0.9 lagging for the two loads.

Step-by-step solution

Step 1 of 7

(a)

Calculate the apparent power:

$$\begin{aligned} P &= S \cos \theta \\ S &= \frac{P}{\cos \theta} \\ &= \frac{2.4 \text{ kW}}{0.8} \\ &= 3000 \text{ VA} \end{aligned}$$

Calculate the power angle.

$$\begin{aligned} \cos \theta &= 0.8 \\ \theta &= \cos^{-1}(0.8) \\ &= 36.86^\circ \end{aligned}$$

Step 2 of 7

Calculate the reactive power:

$$\begin{aligned} Q &= S \sin \theta \\ &= (3000 \text{ VA})(\sin(36.86^\circ)) \\ &= (3000 \text{ VA})(0.6) \\ &= 1800 \text{ VAR} \end{aligned}$$

Calculate the complex power.

$$\begin{aligned} S &= P + jQ \\ &= 2400 \text{ VA} + j1800 \text{ VAR} \\ &= 2.4 \text{ kVA} + j1.8 \text{ kVAR} \end{aligned}$$

Step 3 of 7

Calculate the apparent power for first load.

$$\begin{aligned} S_1 &= \frac{P_1}{\cos \theta_1} \\ &= \frac{1.5 \text{ kW}}{0.707} \\ &= \frac{1500}{0.707} \\ &= 2.121 \text{ kVA} \end{aligned}$$

Calculate the power angle for first load.

$$\begin{aligned} \cos \theta &= 0.707 \\ \theta &= \cos^{-1}(0.707) \\ &= 45^\circ \end{aligned}$$

Step 4 of 7

Calculate the reactive power for first load.

$$\begin{aligned} Q_1 &= S_1 \sin \theta \\ &= (2.121 \text{ kVA})(\sin(45^\circ)) \\ &= (2.121 \text{ kVA})(0.707) \\ &= 1.5 \text{ kVAR} \end{aligned}$$

Calculate the complex power for first load.

$$\begin{aligned} S_1 &= P_1 + jQ_1 \\ &= 1.5 \text{ kVA} + j1.5 \text{ kVAR} \end{aligned}$$

Step 5 of 7

Calculate the complex power for second load.

$$\begin{aligned} S &= S_1 + S_2 \\ S_2 &= S - S_1 \\ &= (2.4 \text{ kVA} + j1.8 \text{ kVAR}) - (1.5 \text{ kVA} + j1.5 \text{ kVAR}) \\ &= 0.9 \text{ kVA} + j0.3 \text{ kVAR} \\ &= 948.68 \angle 18.43^\circ \end{aligned}$$

Calculate the power factor of second load.

$$\begin{aligned} \cos \theta &= \cos 18.43^\circ \\ &= 0.948 \end{aligned}$$

Therefore, the power factor of second load is 0.948.

Step 6 of 7

(b)

Consider the power factor 0.9 (lagging) for the two loads.

$$\cos \theta_2 = 0.9$$

$$\begin{aligned}\theta_2 &= \cos^{-1}(0.9) \\ &= 25.84^\circ\end{aligned}$$

Calculate the parallel element.

$$\begin{aligned}C &= \frac{P(\tan \theta_1 - \tan \theta)}{\omega V_{rms}^2} \\ &= \frac{(2.4 \text{ kW})[\tan(36.86^\circ) - \tan(25.84^\circ)]}{2\pi f V_{rms}^2} \\ &= \frac{(2400)(0.749 - 0.484)}{2\pi(60)(120)^2} \\ &= \frac{636}{2\pi(8.64 \times 10^5)}\end{aligned}$$

Step 7 of 7

Simplify the value of the capacitor further:

$$\begin{aligned}C &= 11.715 \times 10^{-5} \text{ F} \\ &= 117.15 \times 10^{-6} \text{ F} \\ &= 117.15 \mu\text{F}\end{aligned}$$

Therefore, the value of parallel element is $117.15 \mu\text{F}$.